Bioresource Technology 273 (2019) 63–69

Contents lists available at ScienceDirect

Bioresource Technology

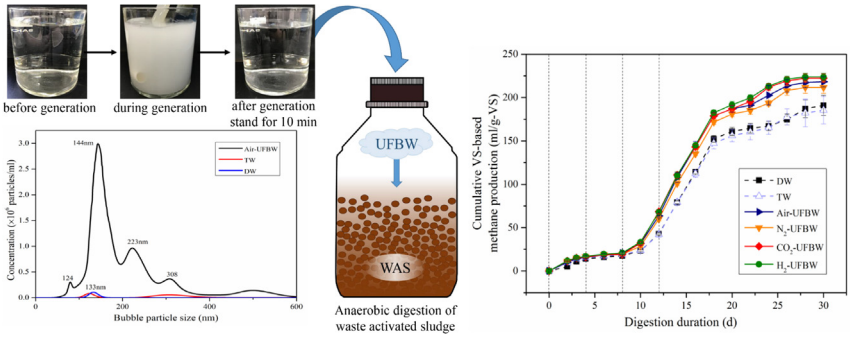
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Characteristics of ultra-fine bubble water and its trials on enhanced methane production from waste activated sludge

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GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Ultra-fine bubble water
Particle size distribution
Methane production
Waste activated sludge
Gompertz model

ABSTRACT

In this study biogas production efficiency was evaluated by adding ultra-fine bubble water (UFBW) into waste activated sludge (WAS) through anaerobic digestion (AD). Four kinds of gases, i.e. Air, N_2, CO_2 and H_2 were introduced into tap water (TW) to prepare the UFBW with their properties being first investigated. Results show that hundreds of millions of nanoscale bubbles with the negative zeta potential could be stable in the UFBW for longer than two weeks whereas almost no nanometer bubbles could be detected in the raw TW. As for their impact on subsequent AD of WAS, the cumulative methane production with the addition of UFBW was 14–21% higher than that from the raw TW addition group. Interestingly, the Air-UFBW also could promote the biogas production in this study, which is different from the common understanding of AD, an obligate anaerobic process.

1. Introduction

Waste activated sludge (WAS) produced from the wastewater treatment plant is rich in organic matters and nutrients that can be recycled and reused, whereas it brings about a lot of serious environmental problems due to its huge production amount, toxic and harmful substances contained and difficult to dispose of (Zhen et al., 2014). Anaerobic digestion (AD) is recognized as the most efficient technique for destroying and stabilizing municipal wastes like WAS, which is basically a microbiological degradation process to convert organic matters under oxygen free conditions into biogas (with methane as the major component), a form of fuel that could reduce handling costs

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https://doi.org/10.1016/j.biortech.2018.10.077
Received 5 September 2018; Received in revised form 26 October 2018; Accepted 27 October 2018
Available online 28 October 2018
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(Feng et al., 2014; Jia et al., 2017; Tampio et al., 2014). AD is becoming a hot topic for sludge treatment due to its energy recovery and cost-effectiveness. However, it has several shortcomings, such as lower hydrolysis rate, longer lag phase, lower biogas production efficiency, etc. (Khan et al., 2016).

In addition to the optimization of operating parameters and enhancement of AD performance via co-digestion, additives addition to the AD system is also an effective way to increase the biogas production (Brown and Li, 2013; Khan et al., 2016). For example, the addition of zero valent iron (ZVI) to AD system has been reported by several researchers. Feng et al. (2014) achieved increased biogas productivity by 43.5% at ZVI of 20 g/L; Jia et al. (2017) pointed out that ZVI could enhance the degradation of chemical oxygen demand (COD) and promote volatile fatty acids (VFAs) generation; and the results from Puyol et al. (2018) indicated that ZVI addition could increase methane production and accelerate the formation of siderite and vivianite. Besides, the addition of ZnO nanomaterials (Zhang et al., 2017), metallic wastes (Montalvo et al., 2018), ferroferric oxide (Yin et al., 2018), CaO_2 (Ping et al., 2018) and bentonite (Ma et al., 2018; Zhang et al., 2018a) has also been reported to promote AD process because of the enhanced microbial hydrolysis, acidification and metabolism under the ambient conditions. In recent years, some researchers have tested the effect of the addition of emerging carbon-based functional materials on AD, including graphene, biochar, activated carbon and carbon cloth due to the key roles of carbons in improving cell growth, enrichment and activity, and accelerating their co-metabolisms (Zhang et al., 2018b). In the research work of Dang et al. (2016), carbon cloth was used as an electron transfer mediator to promote the direct interspecies electron transfer, therefore achieving faster recovery of soured reactors and stimulated methane production than the control. The addition of biochar, on the other hand, might provide temporary substrates to support microbial metabolism and growth that could shorten the lag phase by 21.4% and increase methane production potential by 9.6% (Sunyoto et al., 2016). In addition, ozone was found to initially and temporarily reduce biomass viability and activity, while after the lag phase, methane yield from the ozone-treated sludge could be enhanced by 52% (Chacana et al., 2017). Up to the present, however, most of the above-mentioned methods haven't been applied in practice, probably due to the large amount of chemicals consumption and thus low environmental-friendliness.

Ultra-fine bubble water (UFBW) refers to the water with particle size of bubbles at micro-nano scale and nano-scale. As an emerging technology, it has been a subject of intensive research over the past decade. Although the mechanism is still not yet well understood, it does not prevent the widespread application of UFBW. UFBW has been reported to be effectively applied in many fields such as wastewater or groundwater treatment (Hu and Xia, 2018; Terasaka et al., 2011), industry (Etchepare et al., 2017; Ghadimkhani et al., 2016; Saffari et al., 2013), food and agriculture (Ebina et al., 2013; Liu et al., 2013) and biomedicine (Oeffinger and Wheatley, 2004; Safonov and Khitrin, 2013), most probably owing to their special physicochemical characteristics including stable extended lifetime in water, reduced friction, increased solubility of gases in liquid and charged surface (Terasaka et al., 2011). Still, very little information is available on the microorganisms and its effects on AD performances as well.

To the best of our knowledge, this is the first trial on the effect of UFBW on AD process of WAS. The objective of this research is to explore the characteristics of UFBW produced by introducing different kinds of gases into the UFBW generator and to explore its potential for promoting methane production, aiming to provide a simple and effective method for organics degradation and bioenergy recovery.

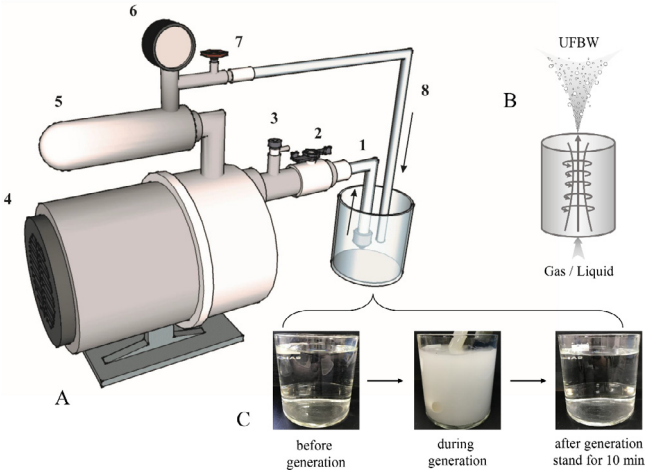


Fig. 1. (A) Schematic diagram of the ultra-fine bubble (UFB) generator. 1. Water inlet, 2. Water inflow valve, 3. Gas inlet (can be connected to the gas bag) and intake valve, 4. Pump, 5. Bubble generating part, 6. Pressure meter, 7. Water outflow valve, 8. Water outlet. (B) Schematic of UFBW generation. (C) Photographs of UFBW before generation, during generation and after generation for 10 min standing.

2. Materials and methods

2.1. The formation and characterization of UFBW

2.1.1. Preparation of UFBW

The combination of pressurized mechanical cyclic, rotational flow and cavitation, an efficient and low-cost approach and widely used in generating ultra-fine bubbles (UFBs) in Japan, was adopted in this study. The schematic structure of UFBs generator (HACK FB11, JAPAN) is shown in Fig. 1A. The liquid was injected into the cylinder together with the specific gas. The liquid in/out flow and gas inflow were manually adjusted to keep the pressure at around 0.25 MPa. A strong shear force produced by increasing pressure was applied on the gas-liquid mixed fluid to make it spin at a high speed. The cavitation could be produced through shearing while pressurizing inside the generation and mixing header, and then bubbles would be generated through a bubbling nozzle (Fig. 1B). In the current experiment, 1.5 L of tap water (TW) in a transparent glass container was recycled into the generator circulation system by respectively introducing Air, N_2, CO_2 and H_2 gases. This generator can produce 8–11 L/min of UFBW generally. Freshly prepared UFBW is a milky white colored opaque liquid, and then become colorless and transparent after standing for 5–10 min (Fig. 1C). This phenomenon may be explained as a large number of visible microscopic bubbles in the bulk water rise to the surface and disappear. The UFBW samples prepared under different generation time (5, 10, 20 and 30 min) and storage durations (0, 7 and 14 days) were used to explore the characteristics (particle size, size distribution, concentration, lifetime and surface charge) of the bubbles in the following experiments.

2.1.2. Bubble size distribution

After UFBW samples being prepared, they were kept in serum bottles for 10 min during which they turned from a turbid to clear liquid. Then it was divided equally (by volume) into the centrifuge tubes and stored in an incubator at 20 °C for further tests. The bubble size distribution was determined by the nano-particle tracking analysis method (MALVERN, NanoSight LM10, UK) at a standard temperature of 20 °C. The bubble size distribution and its concentrations were recorded as a function of storage time.

2.1.3. Dissolved oxygen concentration and pH analysis

The dissolved oxygen (DO) in deionized water (DW), TW and different UFBWs were measured with a DO meter (DKK-TOA DO-31P, JAPAN). The pH value was determined by a pH meter (METTLER TOLEDO FE20, SWITZERLAND).

2.1.4. Zeta potential analysis

Previous research indicates that zeta potential possesses a key role in bubble stability, which is affected by pH (Parmar and Majumder, 2013). In this study, the pH value of prepared UFBW was adjusted from 3.0 to 12.0 by adding 0.1 M NaOH or HCl solution when measuring the zeta potential of UFBW by the Zeta Potential Analyzer (MALVERN, Nano ZS, UK).

2.2. Anaerobic digestion trials

2.2.1. Raw materials

Digested sludge and WAS were sampled from a wastewater treatment plant in Ibaraki (Japan) and stored in the refrigerator (4 °C) prior to the experiment. The water samples used in the experiment were DW (as the control), TW and different UFBWs (including Air-UFBW, N_2-UFBW, CO_2-UFBW and H_2-UFBW).

2.2.2. Experimental design and batch AD tests

Schott Duran serum bottles in two sizes were used in the fermentation experiments. A 1 L bottle was used for preparation and acclimation of inoculum and 250 mL ones were used for methane fermentation. Digested sludge was added into the 1 L reactor and the tested sludge (WAS) was acclimated in this reactor under mesophilic condition (36 ± 2 °C) for about one month. After the above acclimation step, the digestate was used as inoculum. The total solids (TS), volatile solids (VS) of the inoculum and WAS were 1.75%, 1.14% and 3.20%, 2.67%, respectively. In each 250 mL anaerobic digester, 30 mL of inoculum and 70 mL of WAS were added to keep the inoculation ratio (inoculum/WAS, VS basis) in all the reactors around 20%. In this study, a stepwise water addition was applied. 25 mL of DW, TW and four kinds of gas-UFBW addition was performed on days 0, 4, 8 and 12, respectively (totally 4 times during the AD process), and the total volume of added DW, TW or UFBW was 100 mL for each reactor. The initial pH was adjusted to 7.0 ± 0.1 by adding 0.1 M NaOH or HCl solution. Prior to the experimental run, all the reactors were carefully sealed and flushed with high purity nitrogen gas for 2 min to create an anaerobic environment. The reactors were then placed in a temperature-controlled incubator (36 ± 2 °C) till the biogas production stopped (approximately for 30 days). The bottles were shaken once per day at the fixed time prior to sampling the biogas for composition measurement. A 60 mL volume of gas tight syringe was used to measure the daily biogas production by directly reading the scale after being connected to the reactor for 2 min and then normalized to the standard temperature and pressure conditions. One mL of biogas was also sampled and used for biogas composition analysis.

2.2.3. Analytical methods

The contents of TS and VS in the raw and final materials were determined in accordance with standard methods (APHA, 2012). The method of pH determination is the same as in Section *2.3.1* above. Gas composition was determined using a gas chromatography (SHIMAZU GC-8A, JAPAN).

2.2.4. Kinetic models used in this study

The Logistic function is a classical population growth model, which is widely used in the kinetics analysis of anaerobic fermentation because of its good performance in the description of biogas production from AD process (Donoso-Bravo et al., 2010). The following Logistic function (Eq. (1)) was used in this study:

$$P = \frac{P_{max}}{1 + exp\left[\frac{4R_{max}(\lambda - t)}{P_{max}} + 2\right]} \quad (1)$$

where P (ml CH_4/g-VS) is the cumulative CH_4 production at time t, P_{max} (ml CH_4/g-VS) is the maximum CH_4 production potential at the end of incubation time, R_{max} (ml CH_4/g-VS·d) is the maximum CH_4 production rate, t (d) is the duration time and λ (d) is the lag phase defined as a delayed period of a culture in responding to a new environment and starting to produce methane.

Due to the successful modeling of methane production by the modified Gompertz model (Gadhamshetty et al., 2010; Xing et al., 2017), this equation was also used in this study to fit to the experimental data of cumulative CH_4 production under different conditions. The following modified Gompertz equation (Eq. (2)) was employed in this study:

$$P = P_{max} exp\left\{-exp\left[\frac{R_{max}e}{P_{max}}(\lambda - t) + 1\right]\right\} \quad (2)$$

where e is 2.71828, and all the other parameters are the same as in Eq. (1). The correlation coefficient (R^2) was calculated to indicate the kinetic model fitting to the experimental results from methane production.

2.3. Statistical analysis

All the characterizations of bubble properties and fermentation experiments were performed in triplicate. Average value was taken for all the determinations and used for results and discussion. For data analysis, one-way analysis of variance (ANOVA) was used to analyze the statistical differences among the experimental scenarios by using Microsoft Office Excel 2013. Significant difference was assumed at $p < 0.05$.

3. Results and discussion

3.1. Characteristics of UFBW

3.1.1. Determination of UFBW generation time

Four kinds of gases, Air, N_2, CO_2 and H_2, were introduced respectively into TW to produce UFBW at different generation time (5, 10, 20 and 30 min) of the UFBW generator. The results of bubble particle concentration are shown in Fig. 2. Under the condition of 5 and 10 min generation time, almost the similar particle concentrations were detected in the Air-, N_2- and CO_2-UFBW, averagely around 0.5 or

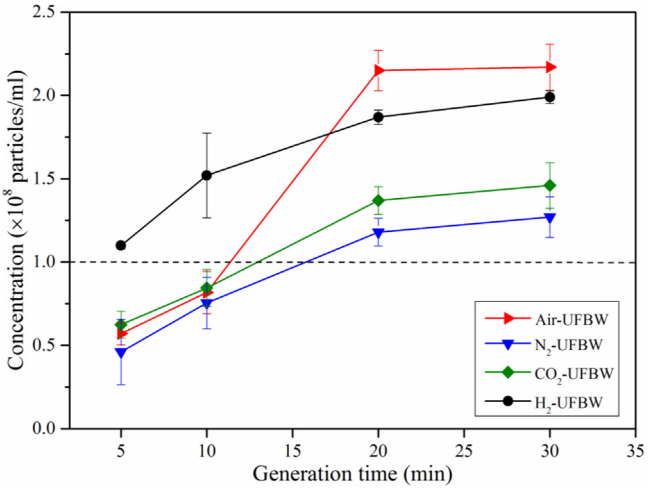


Fig. 2. Particles concentrations in different UFBWs using different gases prepared for different generation time.

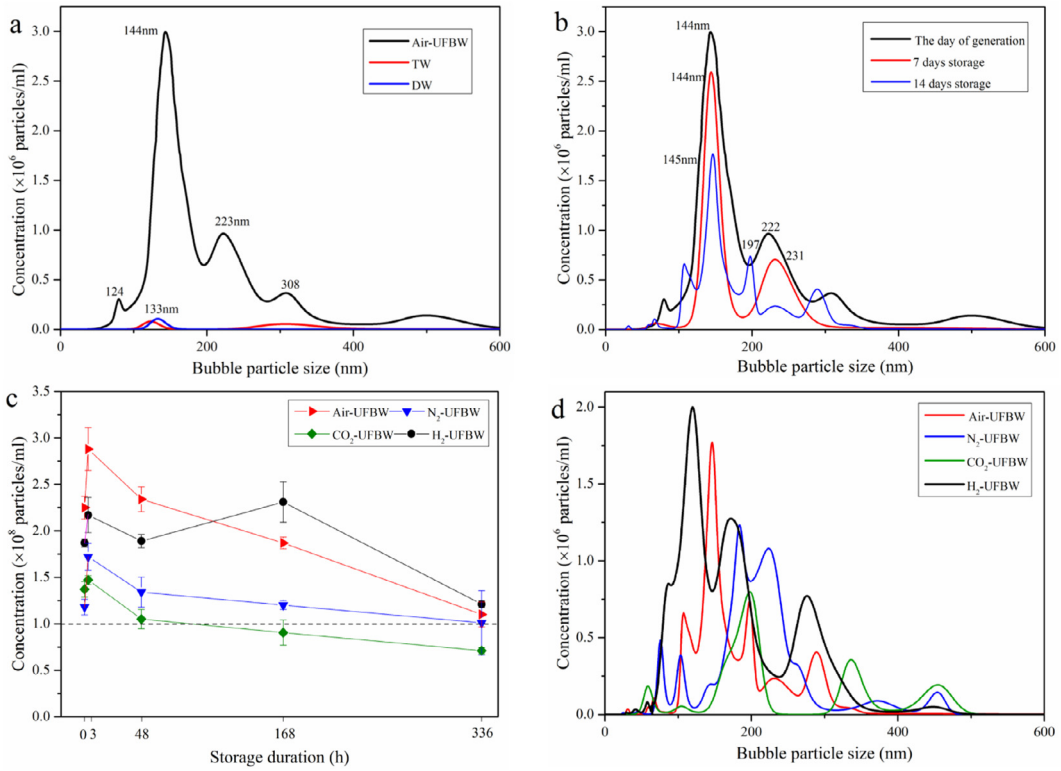


Fig. 3. (a) Comparison of bubble size distribution among the Air-UFBW right after generation, TW and DW. (b) Changes in bubble size distribution in Air-UFBW along with storage time. (c) Variations of particle concentration in different UFBWs after different storage time. (d) Bubble size distribution in different UFBWs after storage for 14 days. UFBW, ultra-fine bubble water; TW, tap water; DW, deionized water.

0.75×10^8 particles/ml, respectively. However, after the generation lasted 20 min, the particle concentration in the Air-UFBW was much higher, almost two times that of N_2- or CO_2-UFBW. For the H_2-UFBW, the bubble concentration exceeded 100 million particles/ml when the generation lasted for 5 min, and then kept increasing with the generation time. In this study, the bubble particle concentration in all the UFBW samples exceeded 100 million particles/ml after the generation lasted for 20 min, much higher than that of the generation for 10 or 5 min. Even though the UFBs concentration increased with the prolonging of generation time, the bubble size and its distribution remained relatively stable when the generation time was longer than 20 min (Fig. 3), therefore 20 min was determined as the generation time for the UFBW production in the following experiments.

3.1.2. Bubble size distribution

A comparison was conducted in Fig. 3 in regards to the bubble size distribution of the UFBW (with Air-UFBW as an example to compare), TW and DW in addition to the change of bubble size distribution and concentration in the different UFBWs after different storage durations. Higher bubble density and more particles in nanometer scale were detected in the UFBW than those in TW or DW, and the major particle size was around 144 and 223 nm in the Air-UFBW right after generation (Fig. 3a). In the case of Air-UFBW, the bubble number density decreased along with the storage duration, from 3.0 to 1.75×10^6 particles/ml with a major particle size around 140 nm. A clear decline tendency in bubble concentration was also observed (Fig. 3b). Interestingly, however, during the 14 days' storage, little change was found in the major particle size (~140 nm) of the UFBs.

Fig. 3c shows that the total bubble number concentrations in the produced UFBW exceeded 100 million particles/ml at the generation day (0–3 h storage), and the highest concentration, 2.88×10^8 particles/ml was achieved in the Air-UFBW, followed by H_2-, N_2- and CO_2-UFBW. With the increase of storage time, the number of bubble particles gradually decreased and dropped to around 1×10^8 particles/ml after storage for 336 h (14 days). Results from Fig. 3c and d show that the bubble number density decreased along with the storage time, either for the concentration of main bubbles or the total number of bubbles. As the results indicated, UFBs could exist in the produced UFBW even after storage for longer than two weeks. After storage for 14 days, the UFBs concentration (which might be indicative of bubble stability) followed a descending order as H_2-UFBW > Air-UFBW > N_2-UFBW > CO_2-UFBW, which may be related to the zeta potentials of the different UFBs (Fig. 4). Among the four kinds of UFBWs, the concentration of bubble particles in the CO_2-UFBs was always the lowest with the shortest lifetime of nanoscale

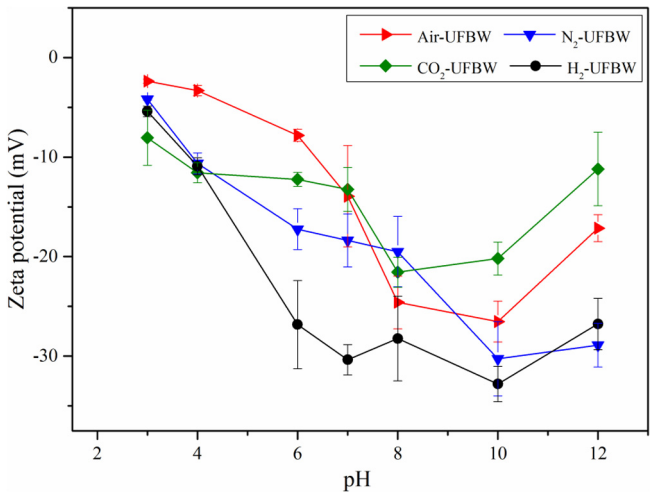


Fig. 4. Changes in zeta potential of UFBs in different UFBWs with variation of solution pH.

bubbles, indicating that the CO_2-UFBs in the bulk water was not so stable compared to other tested UFBs. This observation could be attributable to the stability and solubility of CO_2, which is soluble in water and can combine with water to form carbonic acid in the aqueous solution.

3.1.3. DO concentration and pH value

During the AD process, the stepwise addition of DW, TW and four UFBWs was performed on days 0, 4, 8 and 12, respectively. The usage of air to prepare UFBW did not increase the concentration of DO in the aqueous solution. The DO in the prepared Air-UFBW was 6.32 ± 0.3 mg/L, which is similar to that in DW or TW. Minamikawa et al. (2015) also obtained the similar experimental results. However, when N_2, CO_2 or H_2 was used to prepare UFBW, the DO in the prepared UFBW was reduced significantly. Especially in N_2-UFBW, the DO dropped to 2.86 ± 0.7 mg/L, which is probably due to the degassing effect caused by N_2 gas in the aqueous solution.

As for the pH value in the produced UFBW, since CO_2 is soluble in water, the pH of the CO_2-UFBW was decreased to 5.88 ± 0.4, most probably due to the formation of carbonic acid. The pHs of other UFBWs were basically maintained at neutral with little variation.

3.1.4. Zeta potential of the UFBs

Zeta potential is a key indicator of the stability of colloidal dispersions. A high zeta potential (negative or positive) could create repulsion force that would avoid the coalescence and contribute to the electrical stabilization while colloids with low zeta potentials tend to coagulate or flocculate. In order to further explore the stability of UFBW, zeta potentials of UFBs under different pHs were measured (Fig. 4). Results show that the zeta potentials of UFBs were generally negative at a wide range of tested pH (3–12), which was affected by pH as well as the gas introduced. In a previous report, Ushikubo et al. (2010) pointed out that the zeta potentials of micro- and nano-bubbles were all negative and their absolute value was in the range between 34 and 45 mV (O_2), 17–20 mV (Air), 29–35 mV (N_2) and 20–27 mV (CO_2). The negative value is attributed to the predominance of hydroxide ions in the first molecular layers of water at the gas-liquid interface, thus the different zeta potential value may indicate the different stability of UFBs in the different gas-based UFBWs, and their different structures at the gas-liquid interface as well. In this study, the negative zeta potential increased with the increase of pH value from 3.0 to 10.0 and reached the maximum at pH 10.0 and then decreased when pH was further increased to 12.0. This observation is in agreement with the findings of previous researchers (Calgaroto et al., 2014; Oliveira and Rubio, 2011; Ruckenstein, 2013). In the range of pH from 7.0 to 8.0, which is the most suitable pH condition for methane fermentation, the stability of the different UFBs followed a descending order as H_2 > Air > N_2 > CO_2 in the tested UFBWs with the zeta potentials being higher than −20 mV.

3.2. Effect of UFBW addition on methane production from WAS

The daily and cumulative biogas production as well as the corresponding methane content were determined during the AD process with stepwise addition of DW, TW or different UFBWs (Fig. 5). The AD reactors were run for about 30 days till almost no biogas production being detected. During most of the testing period, the daily biogas or methane production from the AD reactor with UFBW addition was higher than that from the reactor with DW or TW addition. In regard to the daily methane production (Fig. 5c), a peak value of 44.21 mL/g-VS·d was noted on day 14 in the reactor added with Air-UFBW, in comparison to 36.44 and 37.00 mL/g-VS·d from the reactors with DW and TW addition on that day, respectively. While almost no significant difference was detected in daily methane production among the reactors added with different UFBWs. During day 12 to day 14, around 67.16–72.48% and 55.69–66.84% of methane content were detected in the biogas from the UFBW and DW reactors, respectively. This indicated that the addition of UFBW is beneficial for the increase in methane content, possibly resulting from the improved the conversion efficiency. Besides, an average significant difference ($p = 0.023 < 0.05$) in the cumulative methane production was observed between the reactors with UFBW addition and with DW addition. The average cumulative methane yield was obtained in the reactors added with UFBW followed a descending order of 223.88(H_2-UFBW) > 222.35(CO_2-UFBW) > 218.23(Air-UFBW) > 211.58 (N_2-UFBW) ml/g-VS, about 21%, 20%, 18% and 14% higher than that from the reactor with TW addition, and 17%, 16%, 14% and 11% higher than that from the reactor with DW addition, respectively. The relatively higher methane accumulation was obtained from the H_2-, CO_2-, and Air-UFBW added reactors. The addition of N_2-UFBW had slightly less effect, which however also demonstrated some positive effect on methane production in comparison to the control group. This observation may be attributable to the existing nanoscale UFBs in the aqueous solution, which play an important role in AD process. UFBs may affect the physical properties of aqueous environment and improve biological activity, thereby promoting the mass transfer efficiency of nutrients and enhancing the utilization and conversion efficiency of organic substrates. In addition to the effect of these ultra-fine structured bubbles, the introduced gas might also play important roles in methane production. H_2- or CO_2-UFBs in the UFBW might be easily uptaken by the microorganisms for anaerobic fermentation, while N_2-UFBs can ensure a strict anaerobic environment. Most interestingly, the presence of air (or O_2) is generally regarded as toxicant to anaerobic methanogens. However, the Air-UFBW addition can also promote methane fermentation with obviously better effect than DW or TW condition. Probably the presence of limited micro-oxygen could facilitate the hydrolysis of the substrates in the hydrolysis and acidification stages of AD (Xu et al., 2018). The environment could rapidly become anaerobic as the limited oxygen being quickly consumed by aerobic or facultative anaerobic bacteria. On the other hand, some methanogens may have some certain extent of intrinsic tolerance to oxygen exposure, and their bioactivity could be recovered soon after the inhibition of oxygen exposure being removed (Botheju and Bakke, 2011; Zitomer 1998). The synergistic effects of the UFBs as well as the micro-oxygen environment may promote the AD process. This work first proved that Air-UFBW is beneficial for AD process, which still needs further in-depth research to shed light on the mechanisms involved.

3.3. Kinetic analysis of CH_4 production from UFBW added reactors

Under the steady state, the relevant parameters (mean ± standard deviation) obtained by fitting the Logistic and the modified Gompertz models are shown in Table 1. The experimental data well fitted to both Logistic and Gompertz models (R^2 = 0.990–0.995), with the maximum CH_4 production predicted being consistent with the actual experimental results. Compared to the AD reactor added with DW or TW, the reactors with UFBW addition showed relatively higher methane productions at significant difference $p < 0.05$, which could be also indicated by the higher P_{max} and R_{max} value, further confirming that the addition of UFBW is beneficial for methane production. The good simulation results obtained by these two kinetic models in this study reflect that these models can be applied in the actual AD process to predict the biogas production from WAS under UFBW addition.

3.4. Preliminary analysis on mechanisms involved

Results from the multiple parallel experiments showed that the introduction of UFBW enhanced methane production from WAS via AD. Not only H_2- and CO_2-UFBs, but also Air- and N_2-UFBs in water can enhance the efficiency of biogas production. The results suggest that the resultant nanoscale UFBs other than the type of gas play a major role in the AD reactor. Why do UFBs have such effects? One explanation is

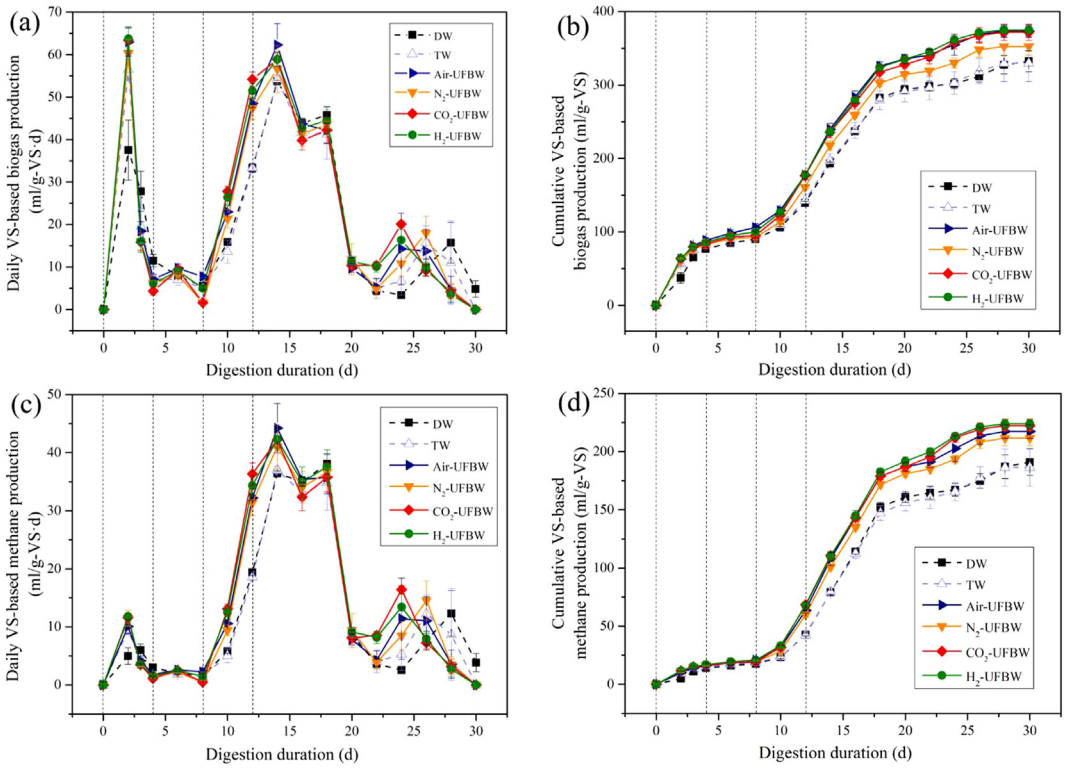


Fig. 5. Daily (a) and cumulative (b) biogas production as well as the corresponding daily (c) and cumulative (d) CH_4 production from the AD trails of WAS with and without UFBW addition. The water addition was applied on days 0, 4, 8 and 12, respectively.

attributable to the changes of the microscale water environment due to the addition of UFBW, like the acceleration of mobility of water molecules in the liquid phase. This could promote the mass transfer of nutrients from the liquid phase to the microbial cells, and thus enhance the bioactivity of microorganism. At a wide range of pH, the zeta potentials of UFBs are negative, which can be explained by the excess of OH^- relative to H^+ ions at the gas-liquid interface, or adsorption of ions in the electrolyte solutions (Demangeat, 2015). The negatively charged UFBs may contribute to the weakening of the hydrogen bonding network in the bulk water. Bubbles can adsorb H^+ ions onto their surface, and the shifting or absence of protons in water could trigger the modification or destruction of the hydrogen bonding network (Chaplin, 2007). Thus for the larger quantity of bulk water that is far away from the surface of UFBs, the hydrogen bonds become weak due to the absence of protons. The weakening of the hydrogen bond network will reduce the water viscosity and accelerate the mobility of water molecules (Bunkin et al., 2007). Also, the formation, shrinkage and disappearance of UFBs would affect the number of water molecules and the structure of water. A previous study (Liu et al., 2013) shows that the addition of nanometer bubbles prolonged the instability time of proton when measured by the proton spin-spin relaxation time, which has been shown to be related to the increase of the mobility of water molecules (Assifaoui et al., 2006; Carini et al., 2016). The increased mobility of water molecules would accelerate the rate of mass transfer of nutrients from liquid to bacterial cells, which further has indirect stimulatory effect on improving the activity of microorganisms and enzymes, facilitating the decomposition of the substrates with improved biogas production. The real mechanisms involved in this complex process need further detailed investigation.

4. Conclusions

This study demonstrated for the first time that with the addition of UFBW, higher methane production can be achieved from the AD

Table 1
Parameters estimated from the Logistic and the modified Gompertz models for methane production from WAS with and without UFBW addition.

Conditions	DW	TW	Air-UFBW	N_2-UFBW	CO_2-UFBW	H_2-UFBW
Logistic model						
P_{max} (ml CH_4/g-VS)	182.92 ± 4.02^{c}	181.32 ± 4.22^{c}	220.87 ± 3.87^{a}	209.03 ± 4.21^{b}	221.35 ± 3.93^{a}	223.68 ± 3.75^{a}
R_{max} (ml CH_4/g-VS·d)	16.78 ± 1.33	15.50 ± 1.20	18.70 ± 1.18	17.82 ± 1.22	18.33 ± 1.07	18.62 ± 1.06
λ (d)	9.21 ± 0.47	8.87 ± 0.49	8.41 ± 0.39	8.48 ± 0.44	8.24 ± 0.39	8.27 ± 0.37
R^2	0.9919	0.9914	0.9945	0.9933	0.9949	0.9954
Gompertz model						
P_{max} (ml CH_4/g-VS)	188.95 ± 6.48^{b}	187.38 ± 7.03^{b}	221.74 ± 6.53^{a}	215.75 ± 6.94^{a}	228.94 ± 6.62^{a}	230.74 ± 6.78^{a}
R_{max} (ml CH_4/g-VS·d)	16.96 ± 1.77	15.90 ± 1.69	19.02 ± 1.68	18.26 ± 1.71	18.65 ± 1.51	19.15 ± 1.61
λ (d)	9.05 ± 0.57	8.79 ± 0.60	8.26 ± 0.50	8.38 ± 0.53	8.08 ± 0.48	8.16 ± 0.49
R^2	0.9902	0.9900	0.9909	0.9904	0.9917	0.9911
P_{max} from the AD trials (ml CH_4/g-VS)	190.93 ± 10.12^{bc}	185.42 ± 11.83^{c}	218.23 ± 5.96^{a}	211.58 ± 6.67^{ab}	222.35 ± 5.14^{a}	223.88 ± 4.21^{a}

Data in the table are expressed as Means ± SD, and the different letters (a, b, c) indicate significant difference at 5% level ($p < 0.05$).

process of WAS. The characteristics of UFBW were detected firstly and the results show that the UFBs in water possess nanoscale particle size, negatively charged surface, strong stability and long lifetime. Both nanoscale UFBs and the introduced gas play important roles in the AD process. Probably the microscale changes of the water environment caused by the addition of UFBW promoted the mass transfer of nutrients to the microbial cells, thereby increasing the efficiency of the methane production.

Acknowledgements

The first author, Ms. Di Wang would like to thank the financial support by the China Scholarship Council for her study in University of Tsukuba, Japan.

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